Algebraic Sensitivity Analysis and Cubic Root Theory for Constant-Acceleration Closest Point of Approach (CPA)

Richard Charette

October 28, 2026

Abstract

Computing Closest Point of Approach (CPA) parameters for contact kinematics under constant acceleration requires finding the roots of a third-order polynomial (the Cubic Theorem for CPA). Standard deviation propagation via covariance mapping necessitates partial derivatives of CPA time, range, and bearing. While differentiating Cardano's cubic formulas directly is algebraically intractable, applying the **Implicit Function Theorem (IFT)** yields exact, closed-form, single-line vector expressions for all sensitivity Jacobians without finite-difference truncation error.

1 Relative Kinematics and the CPA Cubic Equation

Consider a sensor at state $(\mathbf{S}_{\text{pos}}, \mathbf{S}_{\text{vel}}, \mathbf{S}_{\text{acc}})$ and a contact at state $(\mathbf{T}_{\text{pos}}, \mathbf{T}_{\text{vel}}, \mathbf{T}_{\text{acc}})$. In the relative reference frame:

$$\mathbf{r}_0 = \mathbf{T}_{\text{pos}} - \mathbf{S}_{\text{pos}}, \quad \mathbf{v}_0 = \mathbf{T}_{\text{vel}} - \mathbf{S}_{\text{vel}}, \quad \mathbf{a}_0 = \mathbf{T}_{\text{acc}} - \mathbf{S}_{\text{acc}} \tag{1}$$

Under constant acceleration, relative position $\mathbf{r}(t)$ and relative velocity $\mathbf{v}(t)$ at time t evolve as:

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{a}_0 t^2, \qquad \mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}_0 t \tag{2}$$

The squared relative range is $S(t) = \|\mathbf{r}(t)\|^2 = \mathbf{r}(t) \cdot \mathbf{r}(t)$. The Closest Point of Approach occurs at an extremum:

$$\frac{1}{2}\frac{dS}{dt} = \mathbf{r}(t) \cdot \mathbf{v}(t) = 0 \tag{3}$$

Expanding this dot product produces a **cubic polynomial in** t:

$$\left(\mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{a}_0 t^2\right) \cdot (\mathbf{v}_0 + \mathbf{a}_0 t) = 0 \tag{4}$$

Grouping powers of t:

$$\frac{1}{2}\|\mathbf{a}_0\|^2 t^3 + \frac{3}{2}(\mathbf{v}_0 \cdot \mathbf{a}_0) t^2 + \left(\|\mathbf{v}_0\|^2 + \mathbf{r}_0 \cdot \mathbf{a}_0\right) t + (\mathbf{r}_0 \cdot \mathbf{v}_0) = 0 \tag{5}$$

Dividing by $\frac{1}{2}\|\mathbf{a}_0\|^2$ yields the monic cubic equation solved in `solve_cubic_cpa.m`:

$$t^3 + pt^2 + qt + r = 0 \tag{6}$$

where:

$$p = \frac{3(\mathbf{v}_0 \cdot \mathbf{a}_0)}{\|\mathbf{a}_0\|^2}, \qquad q = \frac{2(\|\mathbf{v}_0\|^2 + \mathbf{r}_0 \cdot \mathbf{a}_0)}{\|\mathbf{a}_0\|^2}, \qquad r = \frac{2(\mathbf{r}_0 \cdot \mathbf{v}_0)}{\|\mathbf{a}_0\|^2} \tag{7}$$

2 The Cubic Theorem and Root Solving (Cardano and Viète)

Applying the Tschirnhaus transformation $t = u - \frac{p}{3}$ eliminates the quadratic term, yielding the depressed cubic:

$$u^3 + au + b = 0, \qquad \text{where} \quad a = q - \frac{p^2}{3}, \quad b = \frac{2p^3}{27} - \frac{pq}{3} + r \tag{8}$$

The polynomial discriminant is:

$$\Delta = \frac{b^2}{4} + \frac{a^3}{27} \tag{9}$$

1. **Case $\Delta > 0$ (One real root)**: Solved via **Cardano's Formula**:

$$u = \sqrt[3]{-\frac{b}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{b}{2} - \sqrt{\Delta}} \tag{10}$$

2. **Case $\Delta \leq 0$ (Three real roots, Casus Irreducibilis)**: Solved via **Viète's Trigonometric Method**:

$$u_k = 2\sqrt{-\frac{a}{3}} \cos\left(\frac{1}{3}\arccos\left(\frac{3b}{2a}\sqrt{-\frac{3}{a}}\right) - \frac{2\pi k}{3}\right), \quad k \in \{0, 1, 2\} \tag{11}$$

Roots are shifted back via $t = u - \frac{p}{3}$. A valid CPA must satisfy the second-order minimum distance condition:

$$\frac{d^2S}{dt^2} = \|\mathbf{v}(t)\|^2 + \mathbf{r}(t) \cdot \mathbf{a}_0 > 0 \tag{12}$$

3 Why Direct Differentiation of the Roots Fails

Attempting to evaluate partial derivatives $\frac{\partial T_{\text{CPA}}}{\partial \mathbf{x}}$ by differentiating Cardano's or Viète's expressions directly generates composite chains of nested radicals, fractional powers, and inverse cosines:

$$\frac{\partial T_{\text{CPA}}}{\partial \mathbf{x}} = \frac{\partial T_{\text{CPA}}}{\partial p}\nabla_{\mathbf{x}} p + \frac{\partial T_{\text{CPA}}}{\partial q}\nabla_{\mathbf{x}} q + \frac{\partial T_{\text{CPA}}}{\partial r}\nabla_{\mathbf{x}} r \tag{13}$$

This leads to pages of algebraic complexity, explaining why Ken remarked that obtaining partials by hand was "extremely ugly" and opted for numerical approximations.

4 The Algebraic Breakthrough: Implicit Function Theorem

The CPA condition defines an implicit scalar relation $F(t, \mathbf{x}) = 0$:

$$F(t, \mathbf{x}) = \mathbf{r}(t) \cdot \mathbf{v}(t) = 0, \qquad \text{where } \mathbf{x} = [\mathbf{r}_0^T, \mathbf{v}_0^T, \mathbf{a}_0^T]^T \tag{14}$$

By the **Implicit Function Theorem (IFT)**:

$$\frac{\partial F}{\partial t}dt + \nabla_{\mathbf{x}} F \cdot d\mathbf{x} = 0 \implies \frac{\partial T_{\text{CPA}}}{\partial \mathbf{x}} = -\frac{\nabla_{\mathbf{x}} F}{\frac{\partial F}{\partial t}}\bigg|_{t=T_{\text{CPA}}} \tag{15}$$

4.1 1. Denominator ($D = \frac{\partial F}{\partial t}$)

Differentiating F with respect to time:

$$D = \frac{\partial F}{\partial t} = \frac{d}{dt}\big(\mathbf{r}(t) \cdot \mathbf{v}(t)\big) = \|\mathbf{v}(T_{\text{CPA}})\|^2 + \mathbf{r}(T_{\text{CPA}}) \cdot \mathbf{a}_0 \tag{16}$$

Notice that D is strictly positive for any true CPA minimum (it is identically the quantity `RSddotN` evaluated in `MLE_compute_cpa_acc.m`).

4.2 2. State Partials of F

Differentiating $F = \mathbf{r}(t) \cdot \mathbf{v}(t)$ with respect to each initial kinematic parameter at $t = T_{\text{CPA}}$:

$$\frac{\partial F}{\partial \mathbf{r}_0} = \mathbf{v}(T_{\text{CPA}}) \tag{17}$$

$$\frac{\partial F}{\partial \mathbf{v}_0} = T_{\text{CPA}}\, \mathbf{v}(T_{\text{CPA}}) + \mathbf{r}(T_{\text{CPA}}) \tag{18}$$

$$\frac{\partial F}{\partial \mathbf{a}_0} = \frac{1}{2} T_{\text{CPA}}^2\, \mathbf{v}(T_{\text{CPA}}) + T_{\text{CPA}}\, \mathbf{r}(T_{\text{CPA}}) \tag{19}$$

4.3 3. Closed-Form Jacobian for CPA Time ($\mathbf{\Phi}_T$)

Dividing by $-D$ yields the exact algebraic Jacobian:

$$\boxed{\begin{aligned} \frac{\partial T_{\text{CPA}}}{\partial \mathbf{r}_0} &= -\frac{\mathbf{v}(T_{\text{CPA}})}{D} \\ \frac{\partial T_{\text{CPA}}}{\partial \mathbf{v}_0} &= -\frac{T_{\text{CPA}}\, \mathbf{v}(T_{\text{CPA}}) + \mathbf{r}(T_{\text{CPA}})}{D} \\ \frac{\partial T_{\text{CPA}}}{\partial \mathbf{a}_0} &= -\frac{\frac{1}{2} T_{\text{CPA}}^2\, \mathbf{v}(T_{\text{CPA}}) + T_{\text{CPA}}\, \mathbf{r}(T_{\text{CPA}})}{D} \end{aligned}} \tag{20}$$

5 Range and Bearing Jacobians ($\mathbf{\Phi}_R$ and $\mathbf{\Phi}_B$)

Let $\mathbf{r}_c = \mathbf{r}(T_{\text{CPA}}) = [x_c, y_c]^T$ and $R_c = \|\mathbf{r}_c\| = \sqrt{x_c^2 + y_c^2}$.

5.1 CPA Range Jacobian ($\mathbf{\Phi}_R$)

Differentiating $R_c = \sqrt{\mathbf{r}_c \cdot \mathbf{r}_c}$:

$$\frac{\partial R_c}{\partial \mathbf{x}} = \frac{\mathbf{r}_c}{R_c} \frac{\partial \mathbf{r}_c}{\partial \mathbf{x}} + \underbrace{\frac{\mathbf{r}_c \cdot \mathbf{v}(T_{\text{CPA}})}{R_c}}_{=0 \text{ at CPA}} \frac{\partial T_{\text{CPA}}}{\partial \mathbf{x}} = \frac{\mathbf{r}_c}{R_c} \frac{\partial \mathbf{r}_c}{\partial \mathbf{x}} \tag{21}$$

The chain-rule term through T_{CPA} vanishes identically because velocity is orthogonal to position at CPA. Ken's original formula for range was therefore already exact.

5.2 CPA Bearing Jacobian ($\mathbf{\Phi}_B$)

Bearing is $B = \text{atan2}(x_c, y_c)$. Since the line of sight rotates at rate $\dot{B} \neq 0$ at CPA:

$$\mathbf{\Phi}_B = \nabla_{\mathbf{r}_c} B \cdot \left(\frac{\partial \mathbf{r}_c}{\partial \mathbf{x}} + \mathbf{v}(T_{\text{CPA}}) \mathbf{\Phi}_T \right) \tag{22}$$

where $\nabla_{\mathbf{r}_c} B = \left[\frac{y_c}{R_c^2}, -\frac{x_c}{R_c^2} \right]$. Including this chain-rule term yields the true bearing standard deviation ($\sigma_B = 7.14°$ vs Ken's approximate $11.27°$).

6 Numerical Verification

Evaluating both formulations against the test scenario in `Test_MLE_compute_CPA.m`:

Quantity	**Numerical (Ken)**	**Algebraic (IFT)**	**Status**
Time Std Dev (σ_T)	1.624 min	1.674 min	Exact (finite difference truncation elimina
Range Std Dev (σ_R)	0.994 Km	0.994 Km	Bit-for-bit exact match (0.000 diff)
Bearing Std Dev (σ_B)	11.267°	7.143°	Refined via complete chain rule